NVIDIA RUN:AI · SELF-CERTIFICATION PROGRAM V1.3 × SPECTRO CLOUD PALETTE EDGE

Run:ai Certification — Palette Edge Execution Plan

Certifying the CanvOS / Kairos distribution against NVIDIA Run:ai. Every axis of the certified combination is pinned; every known compatibility gap has an assigned fix.

RUN:AI VERSION	KUBERNETES	DISTRIBUTION
2.23.20	**RKE2 1.34.9**	**CanvOS / Kairos**
control-plane + cluster	edge-rke2 · highest supported by 2.23	edge-native-byoi 2.1.0

`Deploy Edge · AI-RA-Infra-Agent` `Engine RKE2`

`Ingress nginx · Cilium demoted` `GPU DRA · Core-plus-dra` `Kit arch amd64`

TIMELINE · ENGINEERING HOURS

Effort estimate

Focused engineering hours per phase — not wall-clock. The second column is the fast-track figure for this engineer, who reliably lands work in roughly half the standard estimate. External gates (NVIDIA approval, submission review) are not engineering time and don't compress with speed.

NATIVE ESTIMATE	FAST-TRACK (½)	WORKING DAYS @ FAST-TRACK
~102 h	**~51 h**	**~6–7 d**
standard focused effort	this engineer's pace	excl. external gates

P0
Access & prerequisites
4 h fast native 8 h

P1
CanvOS build (Kairos + RKE2)
10 h fast native 20 h

P2
Palette assets & version fixes
12 h fast native 24 h

P3
Provision test cluster
6 h fast native 12 h

P4
Deploy Run:ai (backend → cluster)
5 h fast native 10 h

P5
Pre-flight remediation
5 h fast native 10 h

P6
Run certification kit
6 h fast native 12 h

P7
Evidence & submission
3 h fast native 6 h

Fast-track (½ estimate) Native estimate `bar length ∝ hours`

> **External gates — not engineering hours, unaffected by speed.** Program approval + toolkit/license delivery must precede P0→P1; NVIDIA's review & publish follows P7. Both are wall-clock waits on NVIDIA (days–weeks) and set the true earliest finish regardless of how fast the build goes.

Estimates assume GPU hardware is reserved and Palette tenant access is in place. P2 carries the most effort — the RKE2 layer swap, the nginx ingress cutover (with the single-default-IngressClass fix), three workload-version corrections, and two pack imports all land here.

THE EIGHT PHASES

CanvOS → cluster profiles → evidence

PHASE 1

Build the distribution image

CanvOS `K8S_DISTRIBUTION=rke2`, `K8S_VERSION=1.34.9`, `ARCH=amd64`. Bake Kairos OS + RKE2; push provider image; wire it as `edge-native-byoi`.

EXIT

Provider image + ISO built, pushed, referenced; build SHA recorded.

PHASE 2

Palette assets & fixes

Swap k8s layer to `edge-rke2` 1.34.9; add `AI-RA-Core-Nginx` and demote Cilium ingress; correct Knative / Kubeflow / MPI; define vars; sanitize secrets.

EXIT

Five profiles published, one default IngressClass (nginx), secrets rotated.

PHASE 3

Provision test cluster

Control-plane host + managed GPU cluster on the CanvOS/RKE2 base. Verify Cilium, MetalLB IP, Longhorn, GPU allocatable. Export minimal-privilege kubeconfig.

EXIT

Healthy GPU cluster; kubeconfig in working dir.

PHASE 4

Deploy Run:ai

Backend (control plane) first, then cluster profile — in `install-priority` order. Register the cluster, wire `cluster.uid` + client secret, confirm Connected.

EXIT

Control plane + cluster connected; GPU workload schedules.

PHASE 5

Pre-flight remediation

Confirm nginx ingress honors the Run:ai annotations; verify OIDC/Keycloak login (no 502); confirm all demo secrets replaced.

EXIT

No known-issue blockers; console login verified.

PHASE 6

Run the certification kit

Load `certification-kit-amd64`; run against the cluster with kubeconfig mounted; triage and re-run failures; review the Allure report.

EXIT

Suite complete; results written; failures resolved or documented.

PHASE 7

Evidence & submission

Package `runai-certification-results-<ts>.tar.gz` + Allure report + version manifest + profile exports; submit to NVIDIA.

EXIT

Archive submitted; receipt confirmed.

PHASE 0 · 8

Access & maintenance

P0: toolkit, license, hardware, registry access. P8: re-certify on every new Run:ai GA, K8s, or distro version (quarterly cadence).

EXIT

Toolkit loadable; re-cert log maintained.

RUN:AI 2.23 SUPPORT MATRIX

Compatibility — verified & corrected

Checked against NVIDIA's published Run:ai cluster 2.23 system requirements. Two workload operators in the existing profile fell outside the supported range and one recommended operator was missing — each now has a concrete, tenant-confirmed target.

COMPONENT	2.23 SUPPORTED	IN PROFILE	ACTION
Kubernetes	`1.31–1.34`	`RKE2 1.34.9`	Pinned highest 1.34.x
Knative Serving	`1.11–1.18`	`operator v1.20.0 → Serving 1.20`	`knative-operator 1.18.1` (Serving 1.18)
Kubeflow Training	`1.9.2+ rec.`	`1.8.1`	`kubeflow-training-operator 1.9.3`
GPU Operator	`25.3–25.10`	`25.10.1`	containerd default (RKE2 native)
MPI Operator	`0.6.0+ rec.`	`absent`	Import upstream `mpi-operator ≥ 0.6.0`

Knative 1.18.1 and Kubeflow 1.9.3 are confirmed present in the tenant registries; MPI Operator has no Palette pack and must be imported (upstream `kubeflow/mpi-operator` Helm chart) into a governed registry before it is added to `AI-RA-RunAI-Cluster`.

CERTIFIED STACK

Pinned components

LAYER	COMPONENT	VERSION	PACK	PROFILE
OS	Kairos immutable (BYOI)	2.1.0	edge-native-byoi	AI-RA-Infra-Agent
Kubernetes	RKE2	1.34.9	edge-rke2	AI-RA-Infra-Agent
CNI	Cilium (ingress demoted)	1.18.4	cni-cilium-oss	AI-RA-Infra-Agent
Ingress	ingress-nginx (default)	1.14.3	nginx	AI-RA-Core-Nginx
CSI	Longhorn	1.10.1	csi-longhorn	AI-RA-Infra-Agent
LB	MetalLB (L2)	0.15.3	lb-metallb-helm	AI-RA-Core-plus-dra
GPU	GPU Operator + DRA driver	25.10.1 / 25.8.1	nvidia-gpu-operator-ai · nvidia-dra-driver	AI-RA-Core-plus-dra
Serving	Knative Operator	1.18.1	knative-operator	AI-RA-RunAI-Cluster
Training	Kubeflow Training Operator	1.9.3	kubeflow-training-operator	AI-RA-RunAI-Cluster
MPI	MPI Operator (import)	≥ 0.6.0	mpi-operator	AI-RA-RunAI-Cluster
Run:ai	Control plane	2.23.20	runai-backend / control-plane	AI-RA-RunAI-Backend
Run:ai	Cluster engine	2.23.20	runai-cluster	AI-RA-RunAI-Cluster

Evidence to collect

#	EVIDENCE	SOURCE
1	Certification results archive	results/runai-certification-results-*.tar.gz
2	Allure HTML report	results/allure-report/
3	Version manifest (full triplet + stack)	this plan / kubectl capture
4	CanvOS build metadata (.arg, SHA, image tags)	Phase 1
5	Cluster profile exports (JSON ×5)	Palette API
6	Node / GPU allocatable / operator status	kubectl get nodes -o wide
7	Console screenshot — cluster Connected + GPU	Run:ai console
8	Remediation record (compatibility + secrets)	this plan

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github.com/spectrocloud/CanvOS